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higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

LOGIC SYSTEMS N6

20 NOVEMBER 2019

This marking guideline consists of 9 pages.

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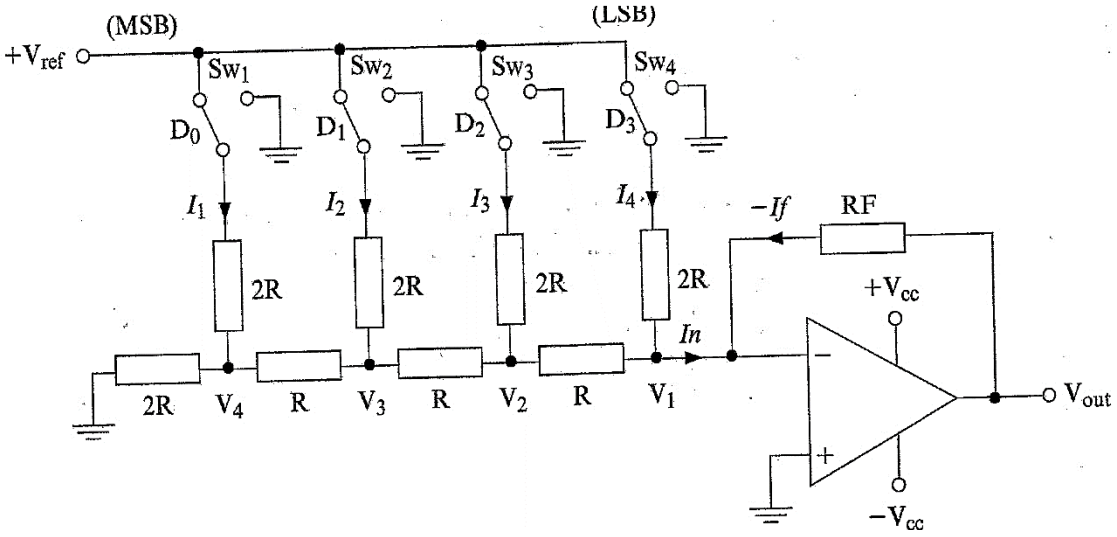
QUESTION 1: OPERATIONAL AMPLIFIERS

1.1 The gain of the inverting operational amplifier is unity (output = input).✓ In this circuit the gain is unity but not inverted and therefore the operational amplifier is faulty and must be replaced.✓✓

INPUT	**×**	**GAIN**	**=**	**OUTPUT**
+6 volts	×	-1 (inverting op-amp)	=	-6 volts✓

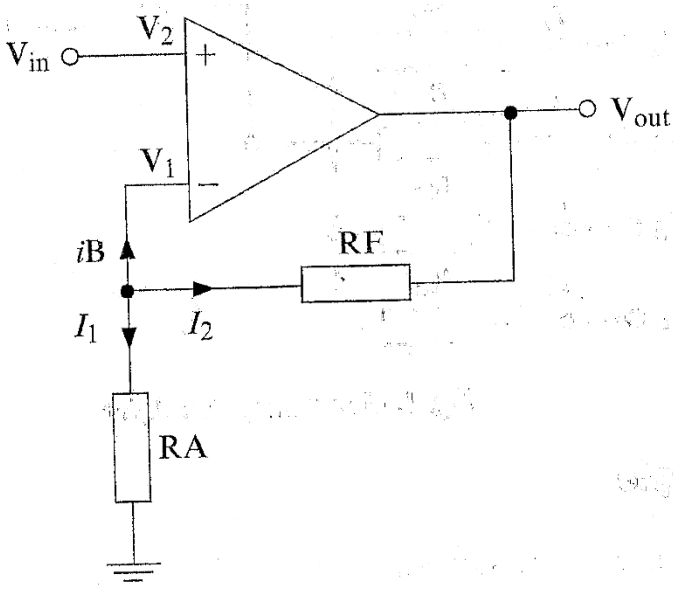
Output = positive which is wrong – it must be inverted to negative.✓ (5)

1.2



(10)

1.3



Mathematical derivation

Proving that the gain is always greater than one:

$I_1 = IF \qquad Vs = I_1 R_1$

$\therefore I_1 = \frac{Vs}{R_1}$

$V_o = I_1 (R_1 + RF)$ ✓

$V_o = \frac{Vs}{R_1} R_1 + RF)$

$= Vs\left(\frac{R_1 + RF}{R_1}\right)$ ✓

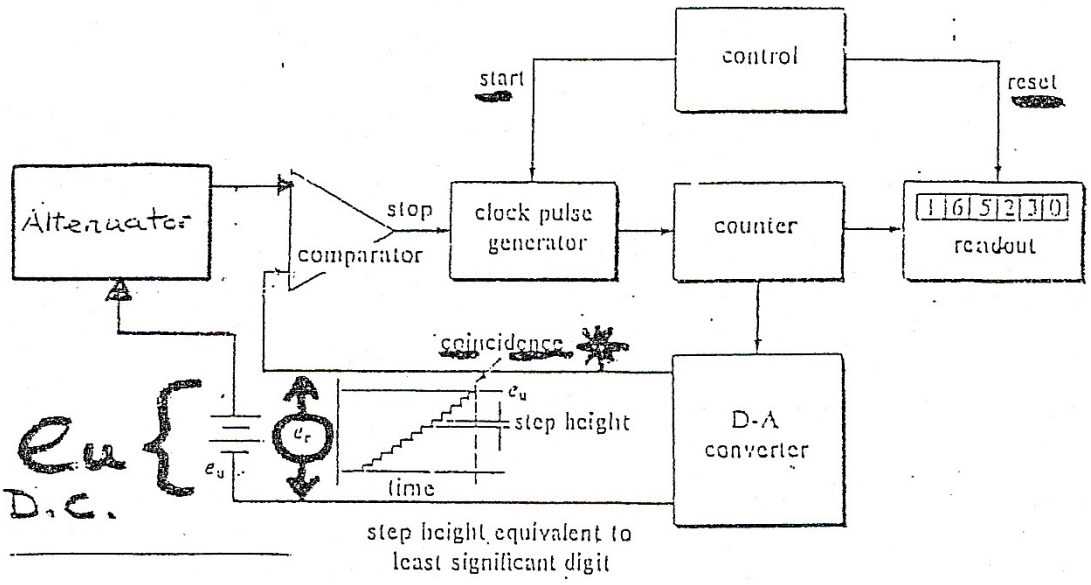
$\therefore V_o = V_s\left(1 + \frac{RF}{R_1}\right)$ ✓

$\therefore V_o > 1$ (5)

[20]

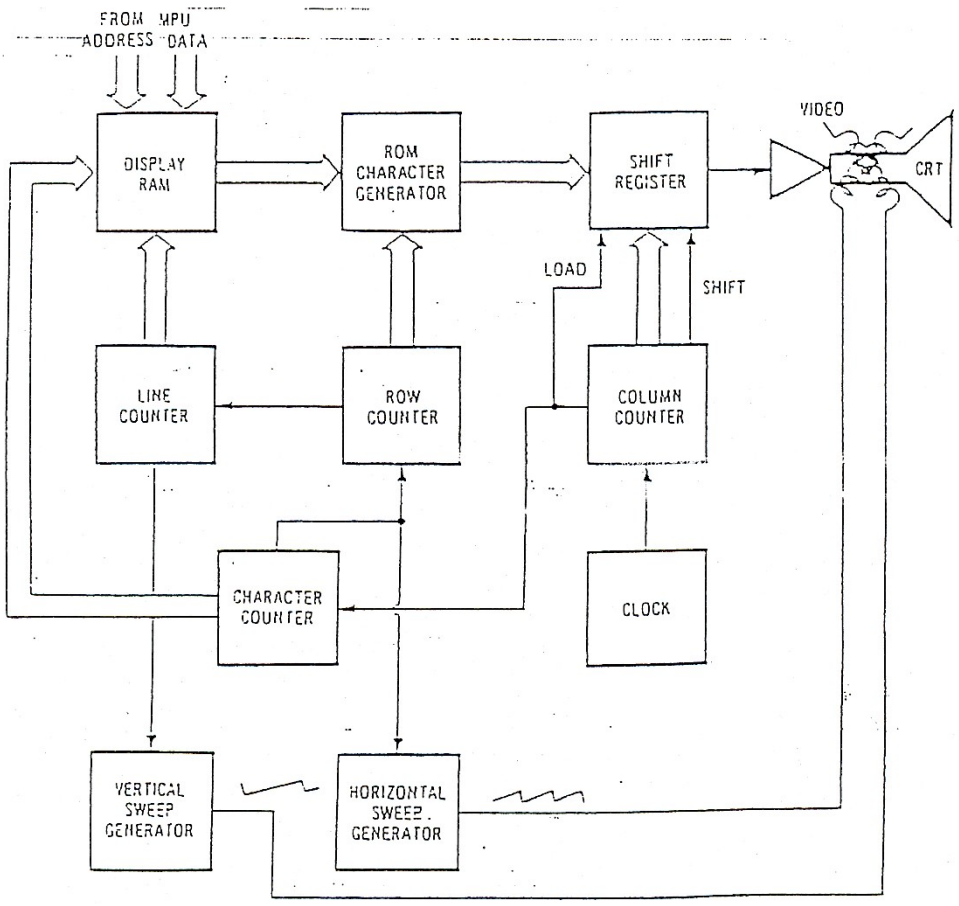
QUESTION 2: INSTRUMENTATION

2.1



(10)

2.2



(Accept other sketches) (10)
[20]

QUESTION 3: TRUE OR FALSE

3.1 True – Logic systems switch you to the cell-transmitter tower with the strongest signal using digital voice communication.

3.2 False – The interface is a port.

3.3 False – OST interfacing involves ACD or DAC. However, an actuator like a stepper motor requires no DAC and some angular displacement transducers give a digital signal direct to the computer.

3.4 False – READ reads RAM.

3.5 False – IR holds OP-code as well.

3.6 False – The program counter holds the address of the next instruction.

3.7 True – The control unit and access to memory and all CPU operations are synchronised by the system-clock oscillator.

3.8 True – Both machine code and assembler are nonportable.

3.9 True – The accumulator stores the output of the ALU.

3.10 True – If there is no JUMP instruction then the program counter will increment: PC + 1.

(10 × 2) **[20]**

QUESTION 4: CIRCUITRY AND INTERFACING

4.1 The trend has been for more integration around microprocessor functions (i.e. digital processing). Hybrid circuits use operational amplifiers (particularly for input functions). Despite the mix-and-match construction designs of the major manufacturers (which blur the classification somewhat) the following can be accepted as a guide to LSI multimeter construction:

4.1.1 Sensing shunting safety – Analogue construction designs and operational amplifiers are used.

Attenuation-amplification filtering – Functions are still op-amp based (i.e. analogue). The trend is to use switching functions from the microprocessor (digital logic switching) to activate these circuits. The resultant mix-and-match of digital and analogue functions are called hybrid circuits. The trend is to extend the range of the microprocessor to take care of all these functions.

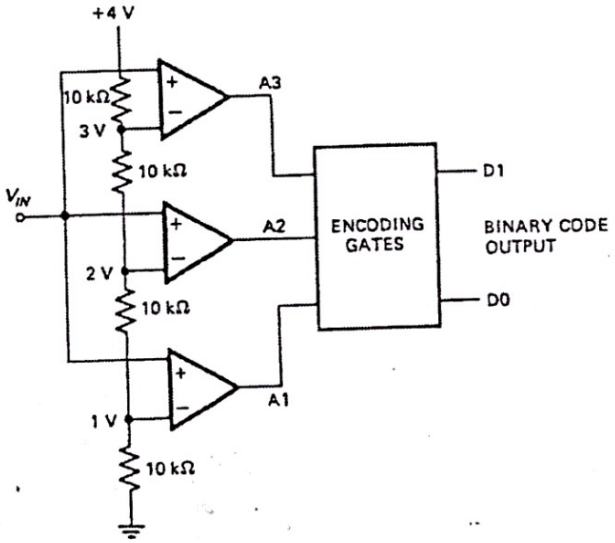
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4.1.2 Processing circuits are based on a microprocessor (digital logic designs). As the power of embedded (i.e. real-time) processing grows, more jobs (i.e. functions) are taken over by the microprocessor, including filtering, amplification and attenuation.

4.1.3 Output circuits are nowadays digitally controlled by the user via choices offered by the microprocessor.

(3 × 2) (6)

4.2



V_{IN} (Volts)	Comparator Outputs			Binary Output	
	A1	A2	A3	D1	D0
0 to 1	0	0	0	0	0
1 to 2	1	0	0	0	1
2 to 3	1	1	0	1	0
3 to 4	1	1	1	1	1

(b)

(Each block = 1 mark✓✓✓✓ table = 2 marks✓✓) (6)
[12]

QUESTION 5: PROGRAMMING INSTRUCTION PROCEDURE

Mathematical operations done before coding:

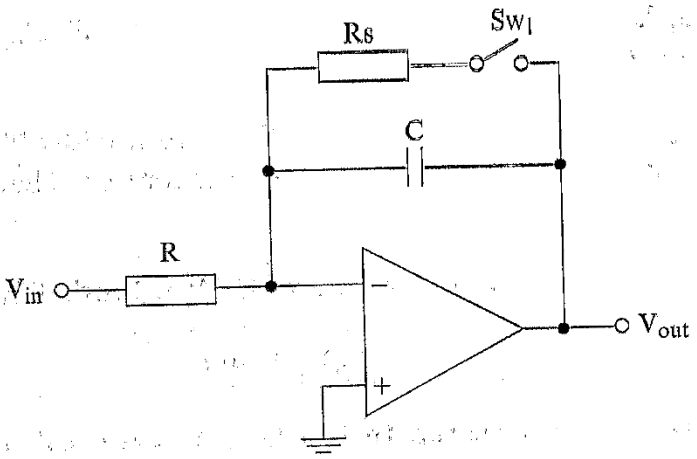
x = ut + uu
x = u(u + t)✓

INSTRUCTION LOCATION	INSTRUCTION WORD		DESCRIPTION
	OP-CODE	ADDRESS	
AAAA	LOAD	F1B4	Load u✓
AAAB	ADDA	F1B3	Add t✓
AAAC	BRAP	AAB2	Branch if acc. >0✓
AAAD	MULT	F1B4	Multiply by u✓
AAAE	BRAP	AAB2	Branch if acc. > 0✓
AAAF	STOR	F1B7	Store in x✓
AAB1	OUTP		Output
AAB2	HALT		Halt✓

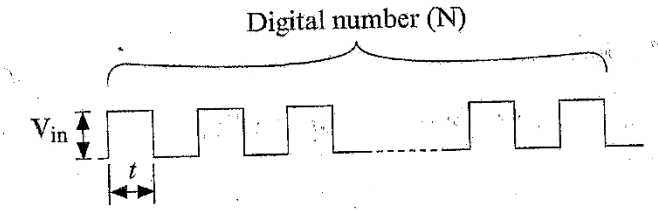
NOTE: Other programming sequences are accepted. Either t or u can be loaded first. **[8]**

QUESTION 6: SIGNAL PROCESSING

6.1 6.1.1



Serial DAC

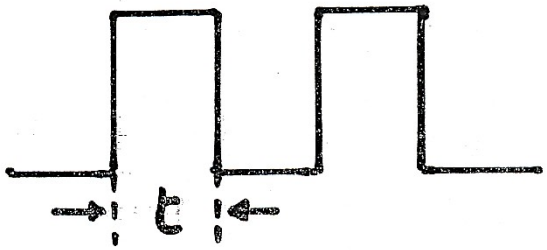


Serial Digital Number

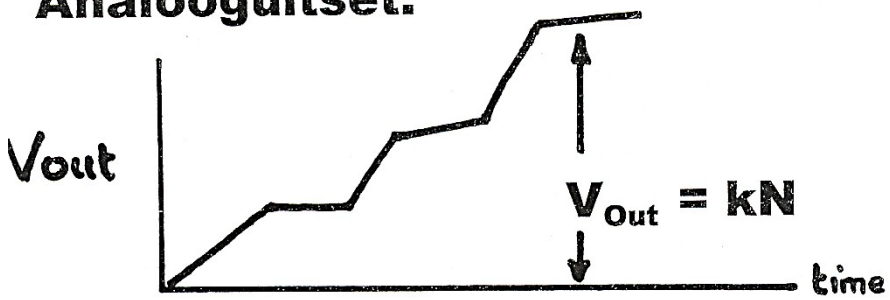
(Each component and waveform = 1 mark) (5)

6.1.2

Digital input:
Syferinset:



Analogue Output:
Analooguitset:

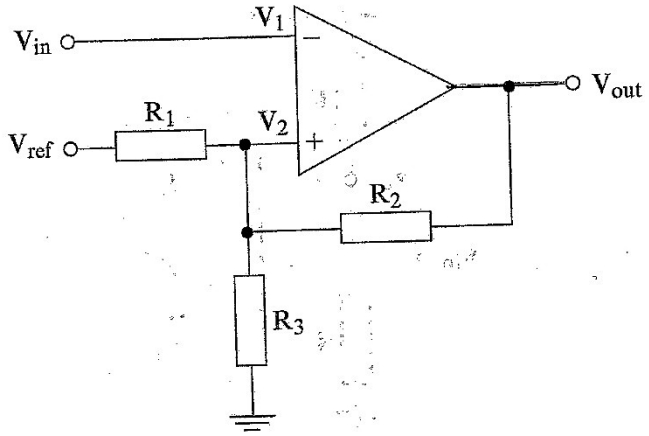


Where k = constants
N = number of Marks

(Each waveform = 1 mark) (2)

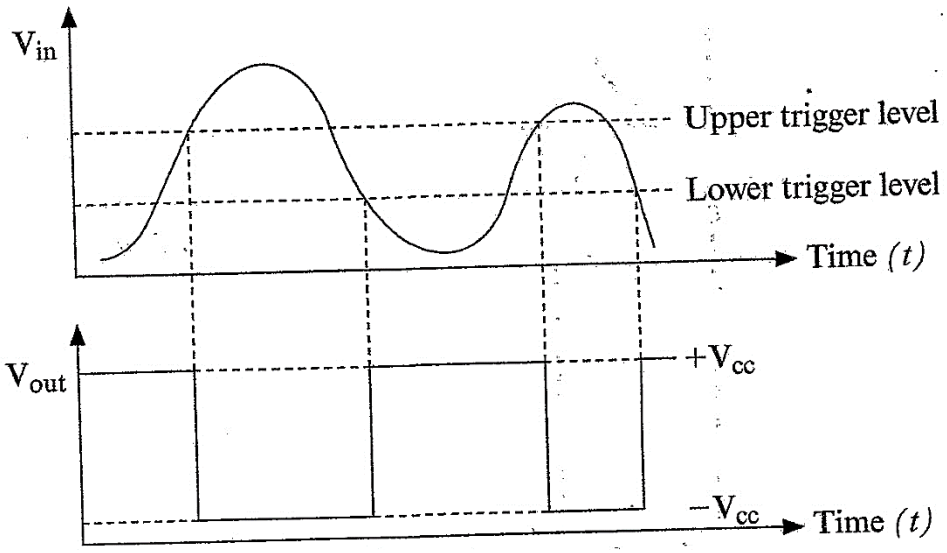
6.1.3
- The voltage in the mark time✓ of the pulse wave form charges the capacitor.✓
- The output voltage is proportional to N (the number of mark pulses) ✓ (3)

6.2 6.2.1



(Each component = 1 mark) (3)

6.2.2



(Each correct waveform = 2 marks) (4)

6.2.3 Schmitt trigger principle of operation:

1. Input is Vin + feedback loop
2. Feedback factor $B=\frac{R_1}{R_1+R_2}$ ✓
3. $V_1 < V$
4. V reference varies
5. $V_1=\left(\frac{R_1}{R_1+R_2}\right) V_R+\beta V_0$ ✓
6. V_1 is lowered to V_2
7. Pulse output✓ (3)

[20]

TOTAL: 100

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